

Total No. of Questions : 8]

SEAT No. :

P3227

[Total No. of Pages : 2

[5354]-683

B.E. (Computer Engineering)
HIGH PERFORMANCE COMPUTING
(2012 Pattern) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) First two questions are compulsory. Answer three questions [(Q.3 or Q. 4), Q.5 or Q. 6) , (Q.7 or Q.8)]*
- 2) Neat diagrams must be drawn whenever necessary*
- 3) Assume suitable data if necessary*

- Q1)** a) What is Dataflow Model? [4]
b) State and explain Basic working principal of Super Scalar Processor?[6]

- Q2)** a) Explain Mapping Technique for load balancing? [6]
b) Write a Short note on Concurrency and Task-Interaction? [4]

- Q3)** a) Explain Principles of Message Passing Programming (MPP)? [8]
b) Describe in details Topologies and Embedding? [7]

OR

- Q4)** a) Explain in detail Non-Blocking Communication Using MPI? [8]
b) Write a Short note on Groups and Communicators? [7]

- Q5)** a) State and explain the Performance Analysis of Parallel Algorithms? [7]
b) Write a short note on Job scheduling? [8]

OR

- Q6)** a) Explain in detail OPENMP: a Standard for Directive Based parallel programming? [8]
b) Write a short note on thread termination? [7]

P.T.O.

- Q7)** a) Explain the Latency hiding/Tolerating techniques and their limitations?[8]
b) Explain in detail 1-D and 2-D Partitioning? [7]
c) Explain Parallel Best-First Search algorithm in detail? [5]

OR

- Q8)** a) Explain in detail Quicksort algorithm with example? [8]
b) Explain the Bandwidth and Latency Limitations? [7]
c) Write a note on 0/1 integer-Linear programming problem with an example? [5]

